

main.py

 Share

Run

```
1 import random
2
3 def insertion_sort_count_shifts(arr):
4     arr = arr.copy()
5     shifts = 0
6
7     for i in range(1, len(arr)):
8         key = arr[i]
9         j = i - 1
10
11         while j >= 0 and arr[j] > key:
12             arr[j + 1] = arr[j]
13             shifts += 1
14             j -= 1
15
16         arr[j + 1] = key
17
18     return arr, shifts
19
20
21 if __name__ == "__main__":
22     log = [18.2, 18.5, 18.9, 17.9, 19.1, 19.4, 19.0]
```

Output

Nearly-sorted shifts: 5
Random shifts: 15
All test cases passed!

=== Code Execution Successful ===